# Bending an ice shelf with meltwater

### A 2-D viscoelastic beam model: derivation and numerics

7 November 2026

This note derives, from scratch, a simple model of how a floating ice shelf bends when a meltwater lake fills (and drains) on its surface, and then shows how to solve it on a computer in a few lines. Every step comes with a picture. The code is in `flexure/viscoelastic_beam.py`. The figures come from `flexure/make_diagrams.py` and `flexure/run_examples.py`.

**The whole model in two lines.** The shelf is a thin Maxwell (spring-and-dashpot) beam floating on sea water. In its simplest form the deflection $w(x,t)$, positive downwards, obeys

$$\left(D\,\frac{\partial^4}{\partial x^4} + k\right)\frac{\partial w}{\partial t} \;=\; \frac{\partial q}{\partial t} \;+\; \frac{q - k\,w}{\tau},$$

where $q = \rho_w g\, d$ is the weight of a lake of depth $d$, $k = \rho_{sw} g$ is the stiffness of buoyancy, $D$ is the flexural rigidity and $\tau$ is the Maxwell relaxation time. When a lake first appears, the ice responds **elastically**. Over a few days it relaxes towards **local isostasy**, $w = q/k$.

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The model actually solved goes one step further. Ice creeps without changing volume, so its viscous Poisson's ratio is $\frac{1}{2}$, while its elastic one is about 0.3. That splits the bending moment into two parts with relaxation times of 4.9 and 3.0 days (§5). The equation above is the special case where the two Poisson's ratios are equal. The split changes deflections by less than 1% and stresses by up to about 20% (in map view).

# 1 Set-up and assumptions

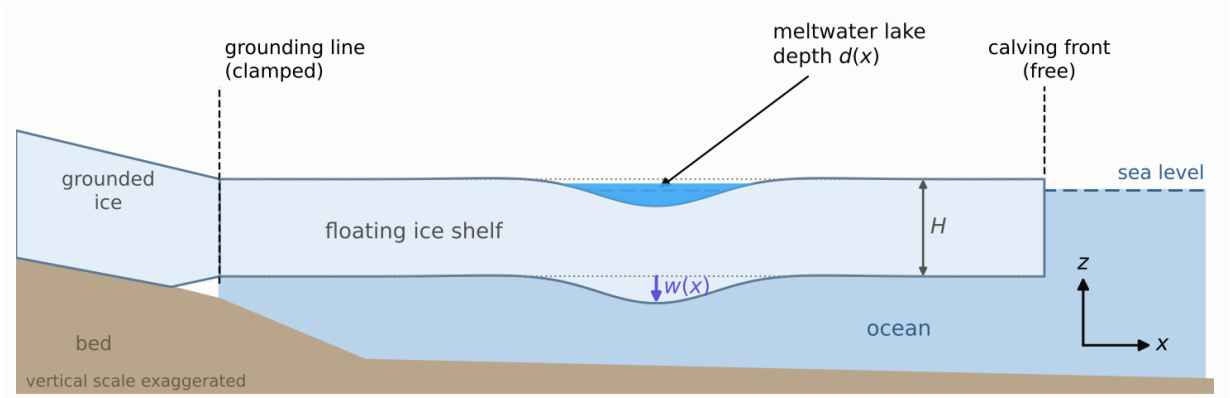



**Figure 1:** A vertical slice through the problem. A meltwater lake of depth $d(x)$ sits on a floating ice shelf of thickness $H$, which deflects downwards by $w(x)$. Vertical scale exaggerated.

We take a vertical slice through the shelf along a line $x$ that runs from the grounding line towards the calving front. $z$ points up. The shelf has thickness $H$, floats in sea water, and carries a lake of water depth $d(x,t)$. The lake's weight pushes the shelf down by $w(x,t)$, measured **positive downwards**. The change in surface elevation is therefore $-w$.

We assume the following:

| assumption | why it is reasonable |
|---|---|
| **2-D, plane strain.** Nothing varies along the third direction $y$, so the lake is a long trough. | It keeps the maths one-dimensional. A circular lake gives similar numbers (see §13). |
| **Thin beam.** Plane cross-sections stay plane. | Lakes are 1 to 5 km across and $H \approx 100$ to 500 m. |
| **Small deflections.** $w \ll H$, small slopes. | $w$ is a few metres. |
| **Linear Maxwell rheology.** | This is the simplest material that is elastic over hours and viscous over weeks. |
| **Hydrostatic ocean.** Sea water pushes back like a bed of springs. | Lakes fill and drain over hours to weeks, much slower than ocean waves. |
| **Prescribed load** $q = \rho_w g d(x,t)$. | We specify the water and do not route it (see §13). |
| **No stretching or flow of the shelf.** | Bending happens much faster than the shelf spreads. |

Everything is linear in the **perturbation**. The background weight of the ice is already balanced by the background buoyancy, so only the extra load $q$ and the extra deflection $w$ appear.

## 2 Buoyancy is a bed of springs

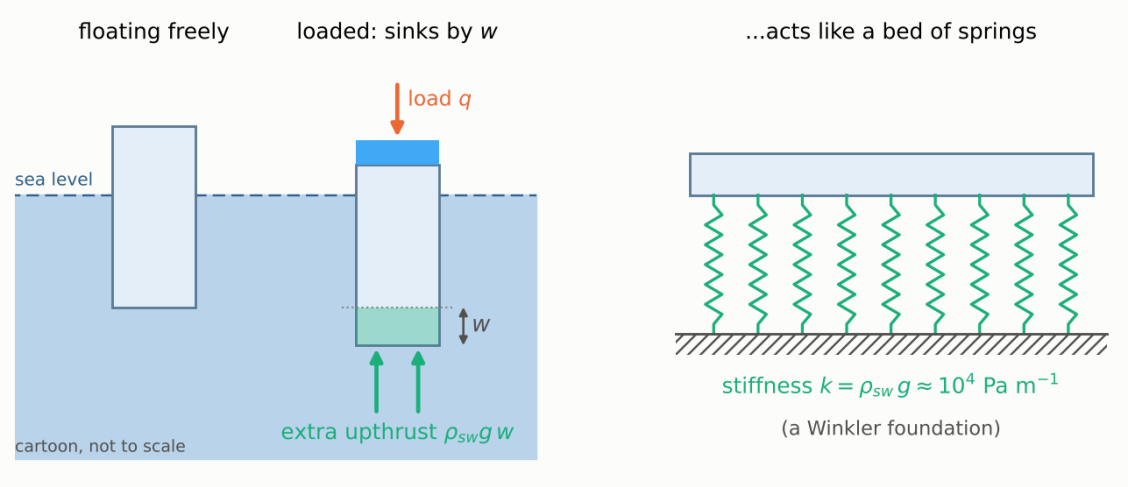



**Figure 2:** Pushing a floating column down by $w$ displaces extra sea water and produces an extra upthrust $\rho_{sw} g\, w$. Buoyancy therefore acts like a bed of springs of stiffness $k = \rho_{sw} g$.

Push a floating column down by $w$ and it displaces an extra $w$ of sea water per unit area. Archimedes then gives an extra upward pressure

$$p_{\text{buoy}} = \rho_{sw}\, g\, w \;\equiv\; k\, w, \qquad k = \rho_{sw} g \approx 1.0 \times 10^4 \text{ Pa m}^{-1}.$$

This acts exactly like a bed of independent springs with stiffness $k$, which is called a *Winkler foundation*. The spring is the ocean, not the ice.

A useful by-product: if nothing else resisted, a lake of depth $d$ would sink the ice by

$$w_{\text{iso}} = \frac{q}{k} = \frac{\rho_w}{\rho_{sw}}\, d \approx 0.97\, d.$$

## 3 Force balance on a slice

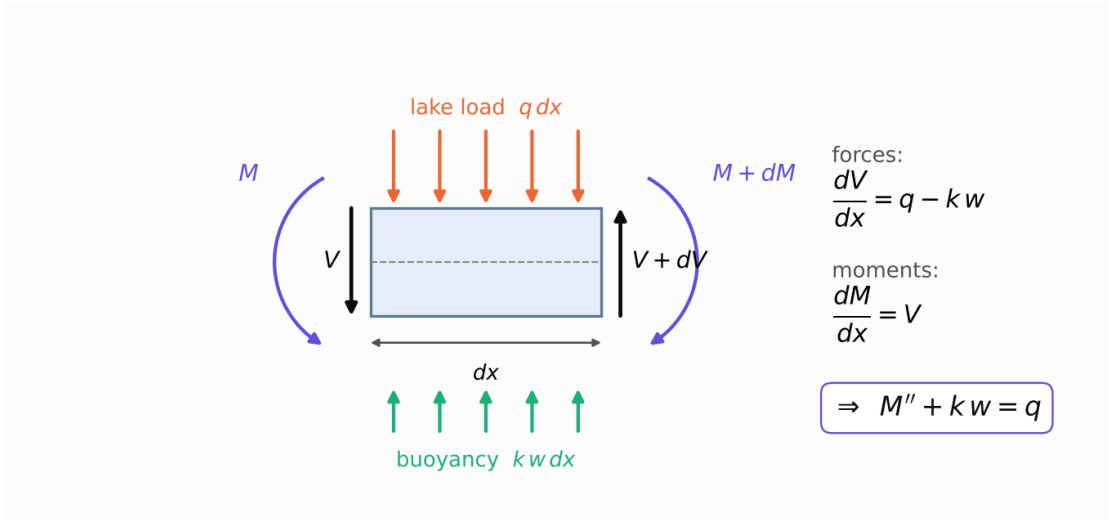



**Figure 3:** Forces and moments on a slice of the shelf of width $dx$.

Cut out a slice of width $dx$. Four things act on it:

- the lake load $q\,dx$, pushing down;
- buoyancy $k\,w\,dx$, pushing up;
- shear forces $V$ and $V + dV$ on its two faces, which come from the neighbouring ice;
- bending moments $M$ and $M + dM$ on its two faces.

**Vertical forces** must balance: $dV = (q - kw)\,dx$. **Moments** must balance (taken about the left face, ignoring tiny $dx^2$ terms): $dM = V\,dx$. Eliminating $V$ gives the equilibrium equation

$$\boxed{\frac{\partial^2 M}{\partial x^2} + k\,w = q} \tag{1}$$

In words: wherever the lake is not held up by buoyancy, the ice must carry the difference through changes in its bending moment.

## 4 Bending: curvature, strain and moment

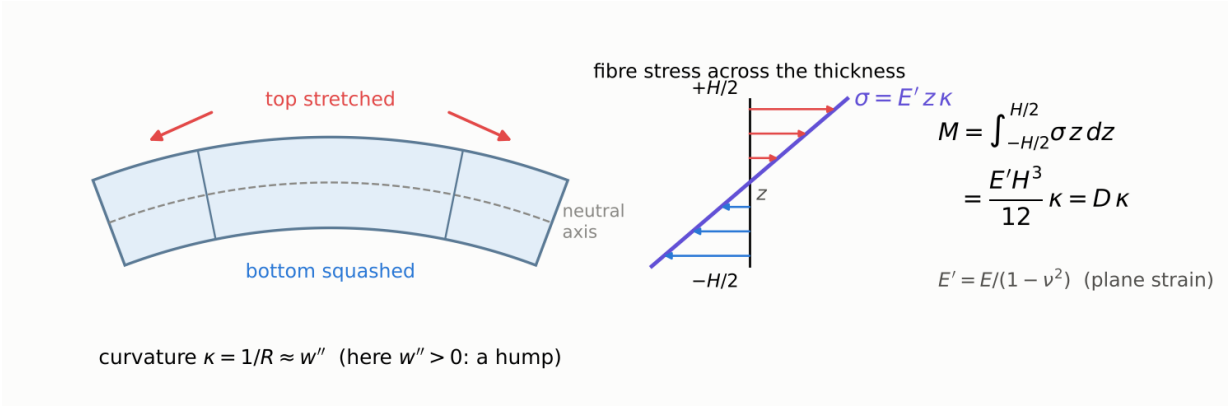



**Figure 4:** Cross-sections stay plane as the beam bends, so fibre strain and stress vary linearly with height $z$. Integrating $\sigma z$ across the thickness gives $M = D\kappa$.

When the beam bends, cross-sections stay straight but tilt. A fibre at height $z$ above the mid-plane (the *neutral axis*) is stretched in proportion to its distance from it:

$$\varepsilon(z) = z\,\kappa, \qquad \kappa = \frac{\partial^2 w}{\partial x^2} \quad \text{(the curvature)}.$$

With $w$ positive down, $\kappa > 0$ means a hump. The top is stretched (tension) and the bottom squashed. Under the middle of a lake the ice sags, so $\kappa < 0$ and the top is in compression.

The **bending moment** is the turning effect of all the fibre stresses $\sigma(z)$:

$$M = \int_{-H/2}^{H/2} \sigma(z)\, z\, dz.$$

For an **elastic** beam in plane strain, $\sigma = E'\varepsilon$ with $E' = E/(1-\nu^2)$. So

$$M = E'\kappa \int_{-H/2}^{H/2} z^2\, dz = \frac{E' H^3}{12}\,\kappa \equiv D\,\kappa, \qquad D = \frac{E H^3}{12(1-\nu^2)}.$$

$D$ is the **flexural rigidity**. It scales as $H^3$, so a shelf twice as thick is eight times as stiff.

For a **viscous** beam, stress depends on strain *rate*. For an incompressible Newtonian fluid of viscosity $\eta$ in plane strain with $\sigma_{zz} \approx 0$, this is $\sigma = 4\eta\,\dot{\varepsilon}$. The same integral gives

$$M = \frac{4\eta H^3}{12}\,\dot{\kappa} \equiv \Lambda\,\dot{\kappa}, \qquad \Lambda = \frac{\eta H^3}{3}.$$

$\Lambda$ is a "viscous flexural rigidity".

## 5 Ice is a Maxwell material

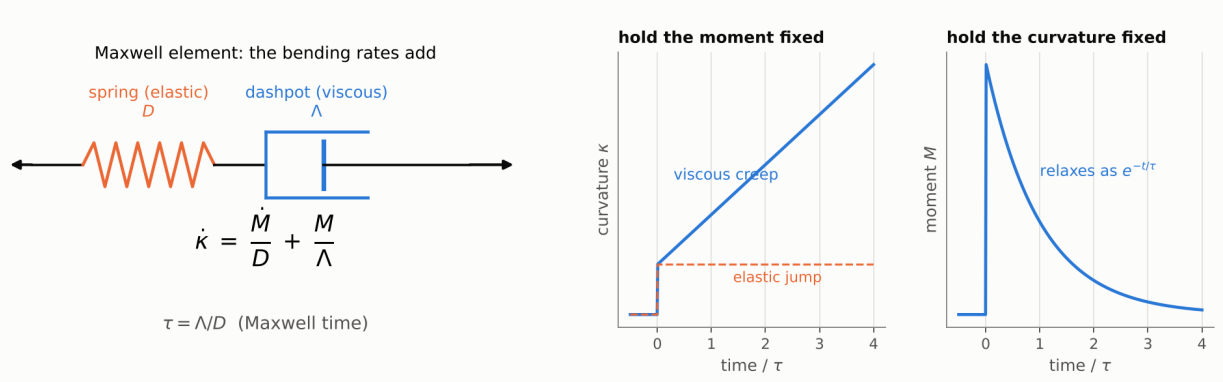



**Figure 5:** A Maxwell element (a spring and a dashpot in series) and its two classic responses: creep under a fixed moment, and relaxation at fixed curvature.

Over minutes to hours, ice behaves like a spring. Over days to weeks it creeps like a very stiff fluid. The simplest model that does both is a **Maxwell element**, a spring and a dashpot in series. Each fibre obeys $\dot{\varepsilon} = \dot{\sigma}/E' + \sigma/4\eta$. Multiply by $z$ and integrate across the thickness, exactly as above, and the *bending* rates simply add:

$$\boxed{\dot{\kappa} = \frac{\dot{M}}{D} + \frac{M}{\Lambda}} \qquad \Longleftrightarrow \qquad \dot{M} = D\,\dot{\kappa} - \frac{M}{\tau}, \tag{2}$$

with the **Maxwell time**

$$\tau = \frac{\Lambda}{D} = \frac{4\eta(1-\nu^2)}{E}.$$

For $\nu \to \frac{1}{2}$ this becomes the familiar $\eta/G$, where $G$ is the shear modulus. With $E = 1$ GPa and $\eta = 10^{14}$ Pa s, $\tau \approx 4.2$ days. The right-hand plots show the two classic experiments:

- **Hold $M$ fixed.** $\kappa$ jumps elastically and then creeps linearly.
- **Hold $\kappa$ fixed.** $M$ jumps and then relaxes as $e^{-t/\tau}$.

A useful mental picture is an **elastic beam whose stress-free shape slowly drifts towards whatever shape it is being held in**.

### 5.1 Two Poisson's ratios, two relaxation times

So far the spring and the dashpot share the same sideways behaviour. They should not. Elastically, ice has a Poisson's ratio $\nu \approx 0.3$. Creep, however, conserves volume, so the viscous part behaves as if its Poisson's ratio were $\nu_v = \frac{1}{2}$ (as in MacAyeal et al., 2021, and Bézu et al., 2025). A single spring–dashpot pair cannot have two Poisson's ratios.

The fix comes from the plate's full moment tensor. Split it into an **isotropic** part (equal bending in every direction) and a **deviatoric** part (more bending one way than the other).

- The elastic stiffnesses of the two parts are $D(1+\nu)$ and $D(1-\nu)$.
- The viscous ones are $\Lambda(1+\nu_v)$ and $\Lambda(1-\nu_v)$.
- Each part is its own Maxwell element.

A beam in plane strain ($\kappa_{yy} = 0$) bends both parts, with weights $(1 \pm \nu)/2$. Its moment is therefore the sum of **two Maxwell elements in parallel**:

$$M = M_1 + M_2, \qquad \dot{M}_i = D_i\,\dot{\kappa} - \frac{M_i}{\tau_i}, \qquad D_{1,2} = D\,\frac{1 \pm \nu}{2}, \tag{2′}$$

$$\tau_1 = \tau\,\frac{1+\nu_v}{1+\nu} = \frac{4\eta(1+\nu_v)(1-\nu)}{E}, \qquad \tau_2 = \tau\,\frac{1-\nu_v}{1-\nu} = \frac{4\eta(1-\nu_v)(1+\nu)}{E}.$$

For $\nu_v = \frac{1}{2}$ these are $\tau_1 = 6\eta(1-\nu)/E \approx 4.9$ days and $\tau_2 = 2\eta(1+\nu)/E \approx 3.0$ days. Four checks:

- **Elastic limit:** $M = (D_1 + D_2)\kappa = D\kappa$, as before.
- **Viscous limit:** $M = (D_1\tau_1 + D_2\tau_2)\dot{\kappa} = \Lambda\dot{\kappa}$, the same viscous rigidity as before.
- **Mean time:** the rigidity-weighted mean $(D_1\tau_1 + D_2\tau_2)/D$ is still $\tau = \Lambda/D \approx 4.2$ days.
- **Equal Poisson's ratios:** if $\nu_v = \nu$ the two times are equal and (2′) collapses back to the single element (2).

In map view the same split gives the viscous plate law $M = \Lambda\big[(1-\nu_v)\dot{\boldsymbol{\kappa}} + \nu_v\,\mathrm{tr}(\dot{\boldsymbol{\kappa}})\,I\big]$. For $\nu_v = \frac{1}{2}$ this is exactly the viscous bending law used by MacAyeal et al. (2021) and Bézu et al. (2025).

### 5.2 Relation to MacAyeal, Sergienko & Banwell (2015)

MacAyeal et al. (2015) wrote down the thin-plate viscoelastic model of ice-shelf flexure under meltwater lakes, motivated by the 2002 break-up of the Larsen B Ice Shelf. Physically, the model in this document is the same one.

**Same framework.** One bending-moment field drives an elastic spring and a viscous dashpot in series, and the plate floats on sea water. Their "buoyancy bucket" is our bed of springs, $k = \rho_{sw} g$.

**Same plate laws.** They split the curvature $\mathbf{H} = (\kappa_{xx}, \kappa_{yy}, \kappa_{xy})$ into elastic and viscous parts, driven by the same moment $\mathbf{M} = (M_{xx}, M_{yy}, M_{xy})$:

$$\dot{\mathbf{H}} = \mathbf{D}^{-1}\dot{\mathbf{M}} + \mathbf{V}^{-1}\mathbf{M}, \qquad \mathbf{D} = D\begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & 1-\nu \end{pmatrix}, \qquad \mathbf{V} = \Lambda\begin{pmatrix} 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}.$$

This is their eqs 17–19. It is written here in our notation and sign convention: their displacement is positive upwards, their viscosity is our $\eta$, and $\Lambda = \eta H^3/3$. **D** is our elastic plate law, and **V** is our viscous plate law with $\nu_v = \frac{1}{2}$.

**Our two relaxation times are their equation, diagonalised.** **D** and **V** share three eigenvectors: isotropic bending $(1,1,0)$, deviatoric bending $(1,-1,0)$ and twist $(0,0,1)$. Their eigenvalues are $D(1+\nu)$, $D(1-\nu)$, $D(1-\nu)$ and $\Lambda(1+\nu_v)$, $\Lambda(1-\nu_v)$, $\Lambda(1-\nu_v)$. In that basis the matrix equation falls apart into independent Maxwell elements, $\dot{M} = d\,\dot{\kappa} - M/\tau$, with

$$\tau_1 = \frac{\Lambda(1+\nu_v)}{D(1+\nu)} \ \text{(isotropic)}, \qquad \tau_2 = \frac{\Lambda(1-\nu_v)}{D(1-\nu)} \ \text{(deviatoric and twist)},$$

which are the times of (2′). Any model with one moment field and two Poisson's ratios therefore has two relaxation times; (2′) just makes them explicit.

**Their experiment, rerun.** They filled a circular lake, 500 m in radius and 2 m deep, on a 200 m thick shelf ($E = 10$ GPa, $\nu = 0.3$) over 60 days, drained it in 6 hours, and followed the shelf for a year. Their ice obeys Glen's law with $n = 3$. The table repeats this with `SpectralPlate` (a 25.6 km periodic box at 50 m, exact in time) for three constant viscosities. It gives the depression at the lake centre; their values are read from their Figs 4 and 5.

| | day 60 (full) | just after drainage | day 365 |
|---|---|---|---|
| MacAyeal et al. (2015), Glen's law $n = 3$ | 58 cm | 36 cm | 19 cm |
| this model, $\eta = 5 \times 10^{13}$ Pa s | 150 cm | 126 cm | 0.4 cm |
| this model, $\eta = 3 \times 10^{14}$ Pa s | 83 cm | 63 cm | 4.0 cm |
| this model, $\eta = 2 \times 10^{15}$ Pa s | 41 cm | 21 cm | 3.5 cm |

The elastic jump when the water leaves agrees. Ours is the purely elastic response to the lake, 19.8 cm, plus up to 4 cm of creep during the 6 hours of drainage when $\eta$ is low; theirs is about 22 cm. The recovery does not agree. With a constant viscosity the centre recovers to within a few centimetres over the following year, and no single value of $\eta$ matches both their filling and their recovery. Under Glen's law the effective viscosity rises as the stresses fall, so their recovery stalls with 19 cm still to go. Their suggestion that repeated fill–drain cycles deepen lake basins year after year therefore rests on the nonlinear rheology.

**Differences.**

- *Rheology.* They offer Glen's law, through an effective viscosity built from the invariants of **M** (their eqs 39–40). Here $\eta$ is constant (see §13).
- *Parameters.* They take $E = 10$ GPa; here $E = 1$ GPa, from tidal flexure. For the same $\eta$ that makes $\ell$ about 1.8 times shorter and $\tau$ 10 times longer.
- *Unknowns and numerics.* They solve for **M**, the viscous curvature and the deflection (seven unknowns) with COMSOL finite elements, and demonstrate one axisymmetric lake with a simply supported edge at $r = 10$ km. Here the 1-D beam uses the scheme of §11. The 2-D plates carry two scalar moment potentials instead of the tensor (see `flexure/viscoelastic_plate.py`), solved by finite differences with grounded ice held fixed, or by FFTs exactly in time.
- *Thickness.* Their tensor form takes spatially varying thickness and viscosity as it stands. The moment potentials need a uniform rigidity in 2-D (see §13).

- *Scope.* They study one idealised lake for one year. Here lakes are routed by fill-spill-merge on real REMA surfaces, with grounding lines, and the routing can be coupled to the flexure.

## 6 The master equation

Now combine the force balance (1) with the material law. Take $\partial^2/\partial x^2$ of each Maxwell law in (2′), add them, and use (1) differentiated in time, $\dot{M}'' = \dot{q} - k\dot{w}$:

$$\dot{q} - k\,\dot{w} \;=\; D\,\frac{\partial^4 \dot{w}}{\partial x^4} \;-\; \frac{\partial^2}{\partial x^2}\Big(\frac{M_1}{\tau_1} + \frac{M_2}{\tau_2}\Big).$$

Rearranged:

$$\boxed{\Big(D\,\frac{\partial^4}{\partial x^4} + k\Big)\frac{\partial w}{\partial t} = \frac{\partial q}{\partial t} + \frac{\partial^2}{\partial x^2}\Big(\frac{M_1}{\tau_1} + \frac{M_2}{\tau_2}\Big)} \tag{3}$$

When the two times are equal ($\nu_v = \nu$), the last term is $M''/\tau$. By (1) that is $(q - kw)/\tau$, and (3) becomes the **single-element master equation**

$$\Big(D\,\frac{\partial^4}{\partial x^4} + k\Big)\frac{\partial w}{\partial t} = \frac{\partial q}{\partial t} + \frac{q - k\,w}{\tau}. \tag{3′}$$

Read (3′) left to right:

- The operator in brackets is the **elastic** beam-on-springs operator.
- $\partial q/\partial t$ is the **instantaneous elastic** response to any change in load.
- $(q - kw)/\tau$ is the **creep** term. Wherever buoyancy is not yet carrying the lake ($kw \neq q$), the ice creeps at a rate set by $\tau$, and it stops only when $kw = q$.

The full equation (3) says the same thing, except that the two parts of the bending moment creep away at slightly different rates.

## 7 Two limits: elastic and isostatic

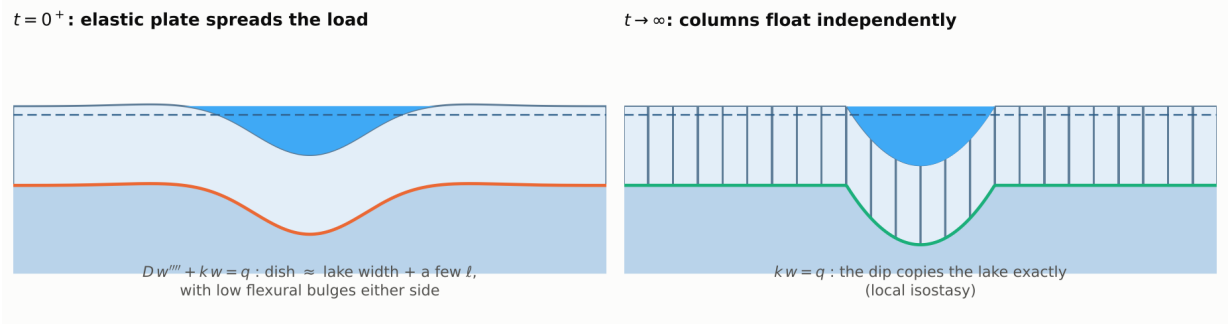



**Figure 6:** The two limits of the model: an elastic response at $t = 0^+$ and local isostasy as $t \to \infty$.

**At $t = 0^+$ the response is elastic.** Integrate (3) across the instant the load appears. The creep term is finite, so it contributes nothing in zero time, and we are left with

$$D\,\frac{\partial^4 w}{\partial x^4} + k\,w = q.$$

This is the classic "beam on elastic foundation" (Hetényi, 1946). Its natural length scale is the **flexural length**

$$\ell = \left(\frac{D}{k}\right)^{1/4} \approx 610 \text{ m} \quad (H = 250 \text{ m},\ E = 1 \text{ GPa}).$$

For a line load $P$ the solution is $w = \frac{P\beta}{2k}\, e^{-\beta|x|}\big(\cos\beta|x| + \sin\beta|x|\big)$, with $\beta = 1/(\sqrt{2}\,\ell)$.

- The dip is spread over a few flexural lengths.
- It crosses zero at $|x| = 3\pi\ell/(2\sqrt{2}) \approx 3.3\ell$.
- A low **flexural bulge** peaks at $|x| = \sqrt{2}\,\pi\,\ell \approx 4.4\ell$, with height $e^{-\pi} \approx 4\%$ of the central dip.

Our 3 km-wide, 4 m-deep lake sinks the centre by 3.2 m at first, instead of 3.9 m.

**As $t \to \infty$ the response is isostatic.** Once nothing changes any more ($\dot{w} = 0$), each Maxwell law makes its $M_i$ decay to zero, so (1) gives $q = kw$. The ice "forgets" its stiffness and floats as independent columns. The dip copies the lake exactly, and the bending moment relaxes to zero.

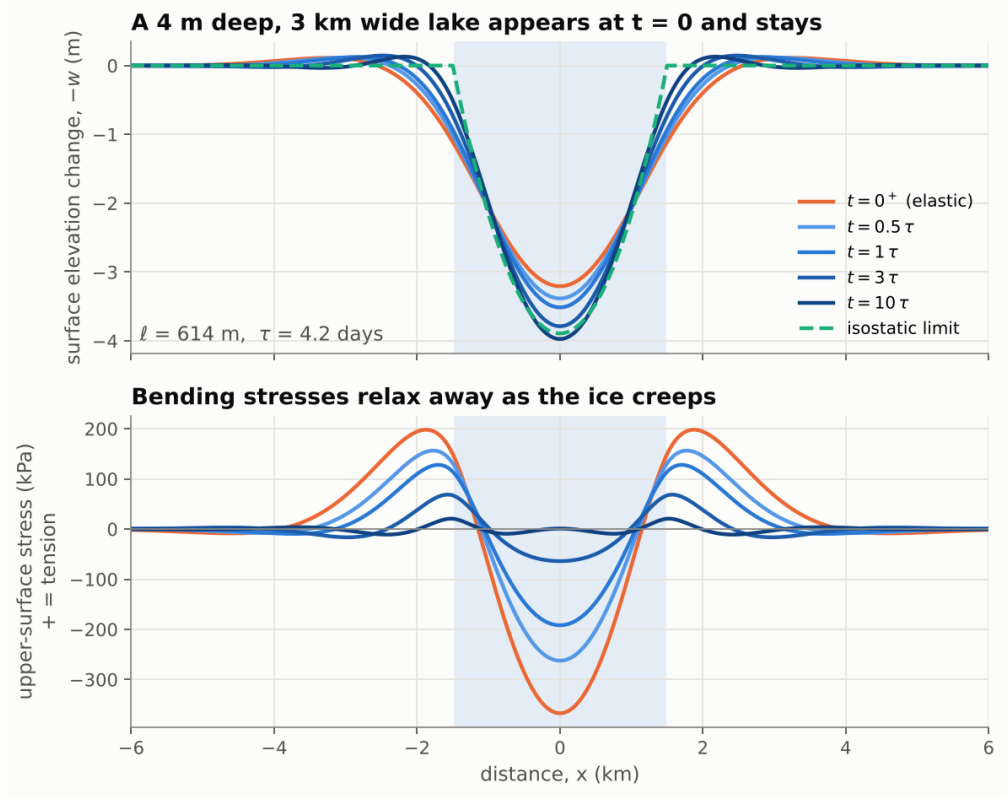



**Figure 7:** A lake appears at $t = 0$ and is left there. The orange curve is the elastic response. The blue curves (light to dark) show the ice creeping towards the dashed isostatic shape. The bending stresses (bottom panel) decay as it does.

## 8 Which shapes relax first? The relaxation spectrum

Because the equations are linear with constant coefficients, each sinusoid in $x$ evolves on its own. Put $w = \hat{w}(t)\cos k_x x$ and $q = \hat{q}\cos k_x x$, with $\hat{q}$ switched on at $t = 0$. Start with the single-element model. Then (3′) becomes an ordinary differential equation with the solution

$$\hat{w}(t) = \underbrace{\frac{\hat{q}}{k}}_{\text{isostatic}} + \Big(\underbrace{\frac{\hat{q}}{Dk_x^4 + k}}_{\text{elastic}} - \frac{\hat{q}}{k}\Big)\, e^{-t/\tau_\lambda}, \qquad \boxed{\tau_\lambda = \tau\big[1 + (k_x\ell)^4\big] = \tau\Big[1 + \Big(\frac{2\pi\ell}{\lambda}\Big)^4\Big]}$$

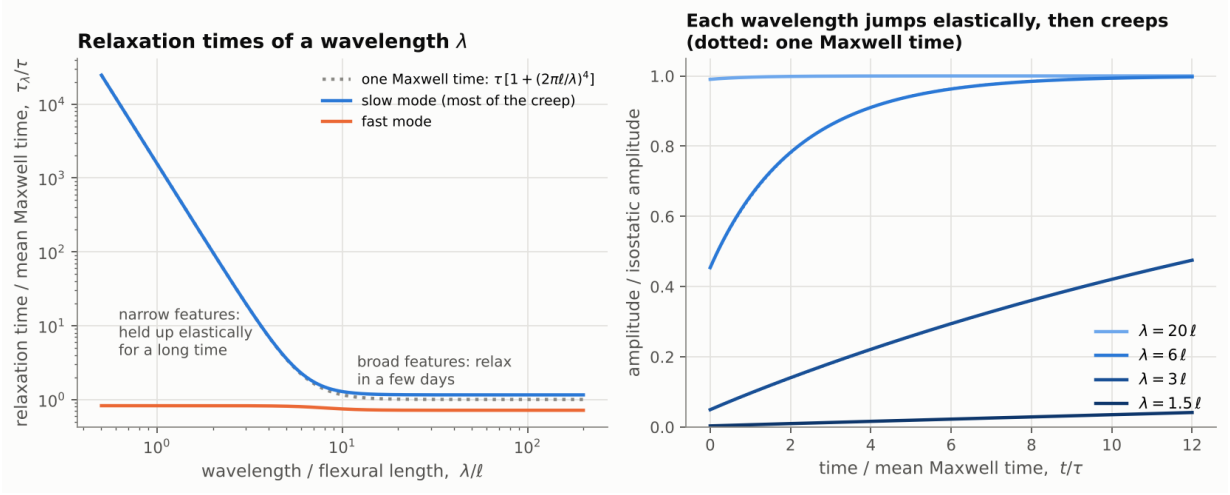



**Figure 8:** Left: relaxation time against wavelength. Right: each wavelength jumps elastically and then relaxes towards isostasy at its own rate.

- **Broad features** ($\lambda \gg 2\pi\ell$) are barely supported by bending in the first place. They are already nearly isostatic, and what is left relaxes in about one Maxwell time.
- **Narrow features** ($\lambda \lesssim 2\pi\ell$) are held up mainly by the ice's stiffness. They relax extremely slowly, as $(2\pi\ell/\lambda)^4$. On a 250 m-thick shelf, a 2 km-wavelength wiggle takes about two months to relax and a 1 km one takes more than two years.

**With two Poisson's ratios** each wavelength relaxes as the sum of two exponentials. Their exact form is a $2 \times 2$ matrix exponential per mode (`relaxation_times` and `modal_step_response` in the code); the left panel above shows both times.

- The **slow mode** stays within 15% of $\tau_\lambda$, and within 1% for narrow features. It carries most of the creep.
- The **fast mode** has a time of 3.0–3.5 days. It carries up to about a third of the (already small) relaxation of broad features, and almost none of the relaxation of narrow ones.

That is why the deflection hardly changes (right panel: dotted lines are the single-element model).

This also tells us that the problem is **not stiff** numerically. The fastest decay rate of any mode is $1/\tau_2$ (about 3 days), so a time step of a fraction of a day is enough for every wavelength.

## 9 Stresses, and why draining is dangerous

The bending stress is largest at the upper surface, where crevasses open:

$$\sigma_{\text{top}} = \frac{6M}{H^2} \qquad (\text{positive} = \text{tension}).$$

For the 4 m-deep, 3 km-wide lake:

- The elastic response puts the surface under the lake centre into compression of about $-370$ kPa.
- A ring of **tension, about $+200$ kPa**, sits just outside the lake shore.
- These stresses are of the same order as those usually needed to open surface crevasses, roughly 100 kPa or more (e.g. Vaughan, 1993).

This is the "ring fracture" mechanism proposed for the 2002 Larsen B break-up (Banwell et al., 2013).

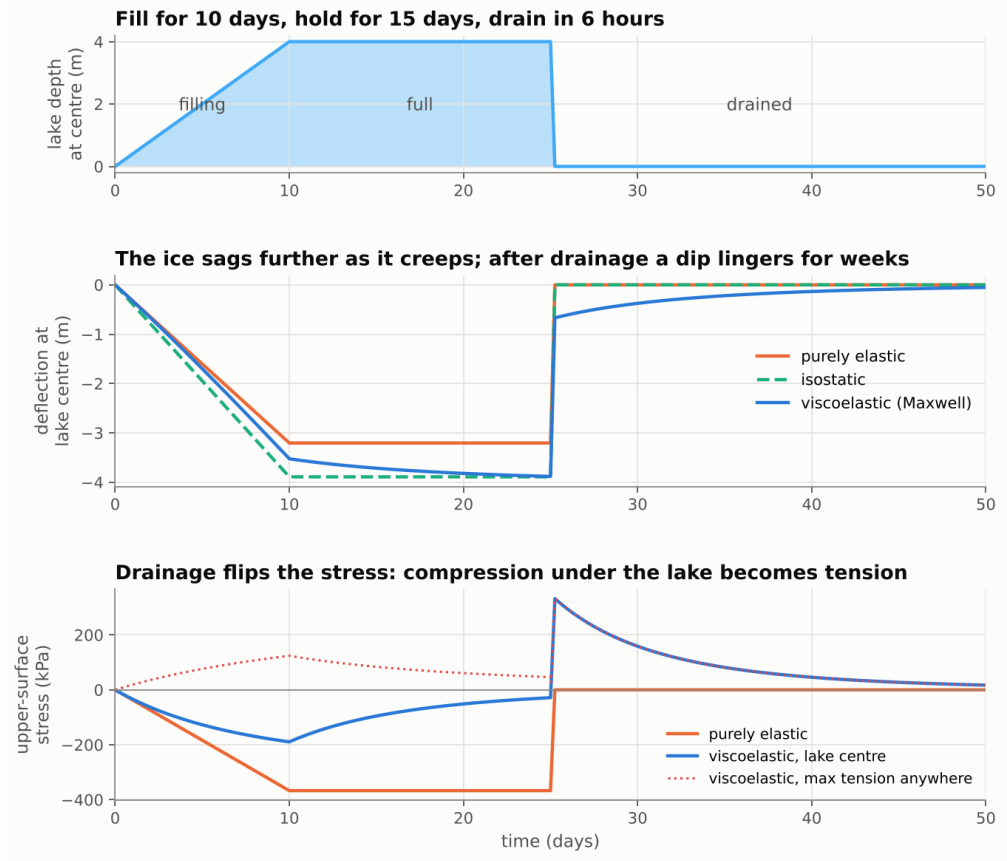



**Figure 9:** A lake fills for 10 days, is held for 15 days and drains in 6 hours. Top: lake depth. Middle: deflection at the lake centre. Bottom: bending stress at the upper surface.

The viscoelastic twist comes with **drainage**:

- **While the lake sits**, the moments relax ($M \to 0$). The ice "gets used to" being bent.
- **When the lake drains** in hours (for example by hydrofracture), the load vanishes much faster than $\tau$. The response to *removing* the load is again elastic, and equal and opposite to the original elastic response. Superposed on a relaxed state, this leaves the ice **bent upwards**, with the surface under the former lake now in **tension (+330 kPa)**.
- A dip of about 0.7 m lingers at the old lake site and creeps away over the following weeks.

Repeated fill-drain cycles therefore reverse the stress at the lake centre again and again. This fatigue-like loading is what the model was built to explore (compare MacAyeal & Sergienko,

2013; Banwell et al., 2019).

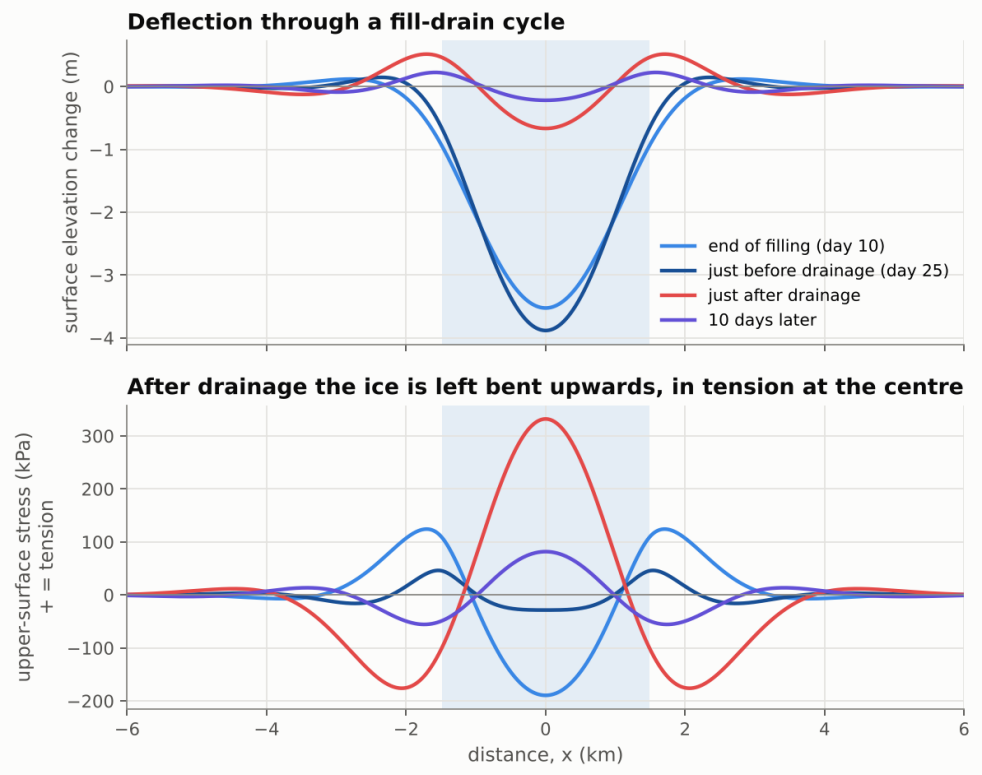



**Figure 10:** Deflection and upper-surface stress at key moments of the fill-drain cycle.

## 10 Boundary conditions

The beam equation is fourth order, so each end needs **two** conditions.

| end | physical meaning | conditions | type |
|---|---|---|---|
| **grounding line** | ice frozen to the bed: no deflection, no tilt | $w = 0,\ \partial w/\partial x = 0$ ("clamped") | kinematic: on $w$ |
| **calving front** | nothing holds the end | $M = 0,\ V = \partial M/\partial x = 0$ ("free") | dynamic: on $M$ |

The free-end conditions are on $M$, not directly on $w$. For a Maxwell beam that starts from rest, $M = 0$ for all time implies $\partial^2 w/\partial x^2 = 0$ for all time, and similarly for the third derivative, so the elastic intuition carries over.

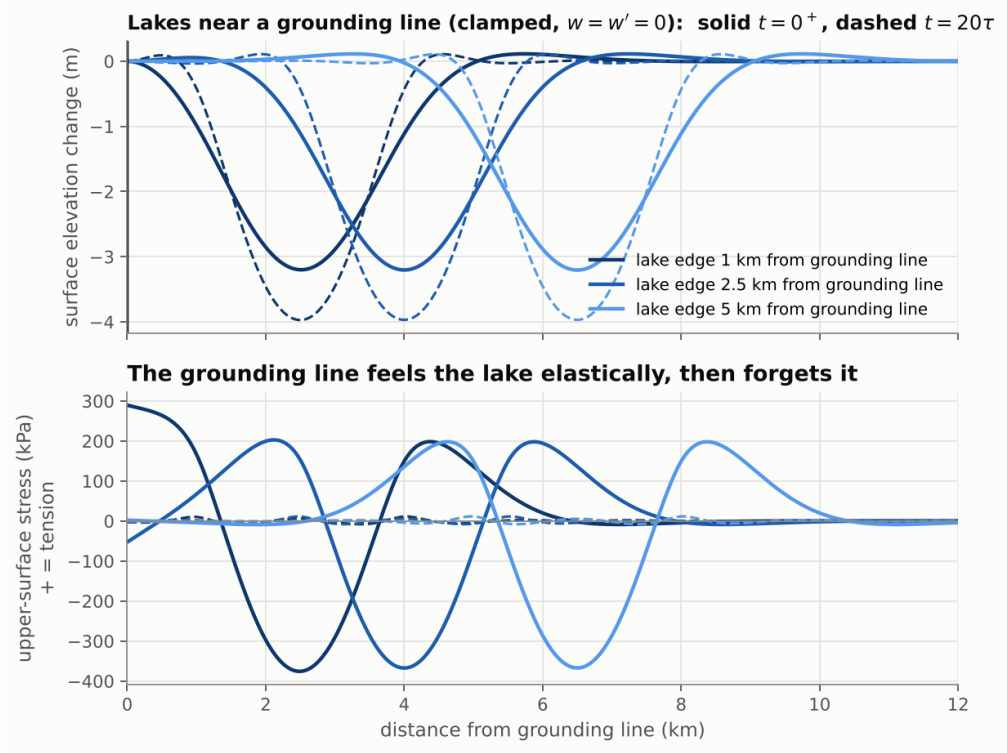



**Figure 11:** Lakes 1, 2.5 and 5 km from a clamped grounding line. A lake within about $2\ell$ of the grounding line strains it elastically: here, +290 kPa of surface tension at the grounding line for the closest lake. That stress relaxes away (dashed curves) as the shelf creeps.

## 11 Numerics: fast and simple

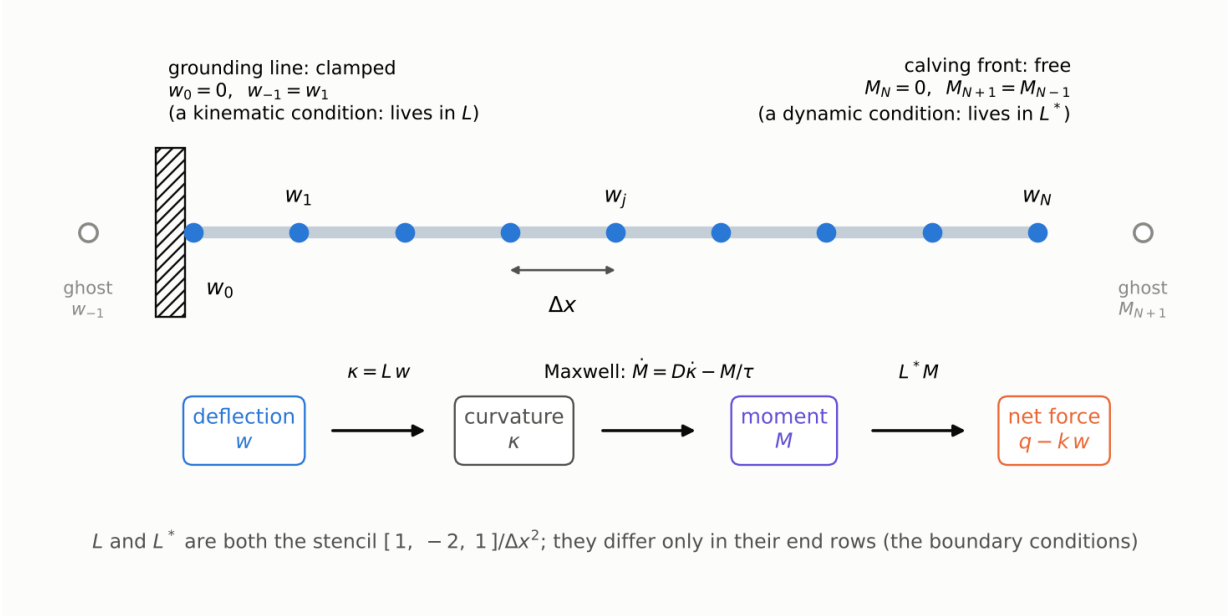



**Figure 12:** The grid, its ghost nodes and the chain of operators $w \to \kappa \to M \to$ force. Kinematic boundary conditions live in $L$ and dynamic ones in $L^*$.

### 11.1 Space: two little matrices

Put $N$ equally spaced nodes $x_j$ along the beam, with spacing $\Delta x$, and store $w_j$ and $M_j$ at each node. The physics is a chain of three steps:

$$w \xrightarrow{L} \kappa \xrightarrow{\text{Maxwell}} M \xrightarrow{L^*} \text{net force } (q - kw).$$

Both $L$ and $L^*$ are the familiar second-difference stencil $[1, -2, 1]/\Delta x^2$. They differ **only in their end rows**, which is where the boundary conditions live:

| end type | row of $L$ (curvature from $w$) | row of $L^*$ (force from $M$) |
|---|---|---|
| clamped (node 0) | ghost $w_{-1} = w_1$: $\kappa_0 = (2w_1 - 2w_0)/\Delta x^2$ | not used: the equation is replaced by $w_0 = 0$ |
| free (node 0) | not used: zeros, so $M_0 \equiv 0$ | ghost $M_{-1} = M_1$: $f_0 = (2M_1 - 2M_0)/\Delta x^2$ |

Kinematic conditions go in $L$ and dynamic ones go in $L^*$. The fourth derivative is never written down explicitly. It is just $L^* D\, L$, so a spatially varying $D(x)$ (a thinning shelf) costs nothing extra.

### 11.2 Time: the trapezoidal rule, then one linear solve

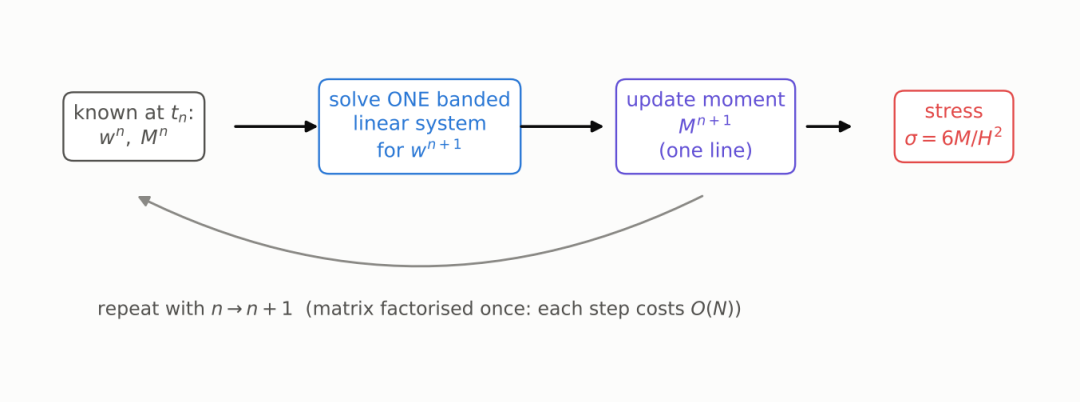



**Figure 13:** One time step of the algorithm.

Step from $t_n$ to $t_{n+1} = t_n + \Delta t$. For each part $i = 1, 2$ write $a_i = \Delta t/\tau_i$ and $c_i = 1/(1 + a_i/2)$.

**1. Moment update.** Apply the trapezoidal (Crank–Nicolson) rule to each Maxwell law $\dot{M}_i = D_i L\dot{w} - M_i/\tau_i$ at every node:

$$M_i^{n+1} = c_i \left[ (1 - \tfrac{a_i}{2})\, M_i^n + D_i\, L\, (w^{n+1} - w^n) \right].$$

**2. Deflection solve.** Substitute these into the force balance at the new time, $L^*(M_1^{n+1} + M_2^{n+1}) + k\, w^{n+1} = q^{n+1}$. This gives **one linear system** for the new deflection, with $\bar{D} = c_1 D_1 + c_2 D_2$:

$$\Big( \underbrace{L^* \bar{D}\, L + k\, I}_{\text{constant, pentadiagonal}} \Big)\, w^{n+1} = q^{n+1} + L^* \bar{D}\, L\, w^n - L^* \Big[ \textstyle\sum_i c_i\, (1 - \frac{a_i}{2})\, M_i^n \Big].$$

**3. Repeat.** Solve step 2 for $w^{n+1}$, then evaluate step 1 for each $M_i^{n+1}$.

That is the entire algorithm. In the code it reads (`Mp0` holds the two parts, and `c`, `a` have one row per part):

```
def step(self, w0, Mp0, q1, dt):
    lu, S, c, a = self._step_matrices(dt)          # S = L* (c1 D1 + c2 D2) L, factorised on
    rhs = q1 + S @ w0 - self.Lstar @ (c * (1 - a/2) * Mp0).sum(axis=0)
    w1 = self._solve(lu, rhs)                       # one banded back-substitution
    Mp1 = c * ((1 - a/2) * Mp0 + self.Dp * (self.L @ (w1 - w0)))
    return w1, Mp1
```

### 11.3 Why this is a good scheme

- **Fast.** The matrix never changes, so it is factorised once. Each step is then one banded back-substitution costing $O(N)$. 3000 nodes × 1200 steps take about one second in pure Python and SciPy.
- **Exact elastic limit.** With $\Delta t/\tau \to 0$ it reduces to the elastic solve, so instantaneous loading and unloading are captured exactly.
- **Second-order accurate** in both $\Delta x$ and $\Delta t$, and **unconditionally stable**: the trapezoidal rule is A-stable.
- **Force balance holds to round-off at every step.** The two operators are used consistently, so the discrete version of (1) is satisfied exactly, and the stresses are physically consistent.
- **Choosing steps.** $\Delta x \lesssim \ell/10$ and $\Delta t \lesssim \tau_2/10$ give errors well below 0.1 % (see the convergence panel below).

## 12 Verification

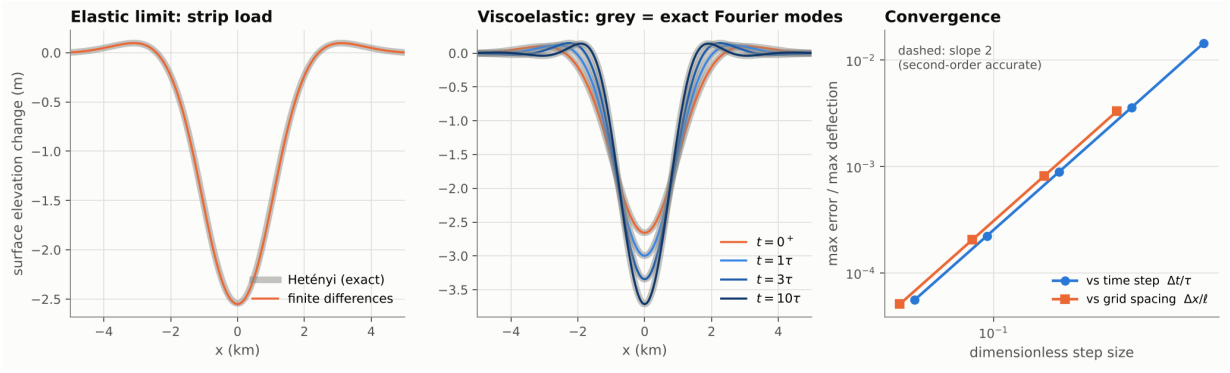



**Figure 14:** Verification. Left: elastic strip load against Hetényi's exact solution. Centre: viscoelastic relaxation against the exact Fourier-mode solution (grey). Right: second-order convergence in $\Delta t$ and $\Delta x$.

The test-suite (`flexure/tests/test_viscoelastic_beam.py`, run with `python -m pytest -q`) checks the model against exact solutions:

| test | exact solution | agreement |
|---|---|---|
| elastic strip load, free beam | Hetényi (1946) | < 0.2 % |
| line load at a free calving front | $w = \frac{2P\beta}{k} e^{-\beta x}\cos\beta x$ | < 0.2 % |

| test | exact solution | agreement |
|---|---|---|
| uniform load, clamped grounding line | $w = \frac{q}{k}\left[1 - e^{-\beta x}(\cos\beta x + \sin\beta x)\right]$ (the tidal-flexure profile) | < 0.1 % |
| viscoelastic relaxation of a lake | exact Fourier-mode solution (§8, two-element) | < 0.2 % |
| equal Poisson's ratios | the single-element closed form $\tau_\lambda = \tau[1 + (k\ell)^4]$ | to round-off |
| moment tensor in map view | $D[(1-\nu)\kappa + \nu\,\mathrm{tr}\,\kappa\, I]$ elastically, $\Lambda[(1-\nu_v)\dot\kappa + \nu_v\,\mathrm{tr}\,\dot\kappa\, I]$ when viscous | $< 10^{-4}$ |
| long-time limit | local isostasy $w = q/k$, $M \to 0$ | ✓ |
| uniform load on a free beam | no bending at all | to round-off |
| discrete force balance $L^*M + kw = q$ | (all boundary-condition combinations) | to round-off |
| time convergence | slope 2 | ✓ |

## 13 Limitations and easy extensions

- **Lake–flexure feedback.** In this document the water depth is prescribed. In reality the lake sits in the depression it creates. The routing notebooks couple the plate to fill-spill-merge, so water and ice shape each other:
  - `flexure/lake_drainage_flexure.ipynb` drains a routed lake;
  - `flexure/coupled_routing_flexure.ipynb` routes a melt season over the flexing shelf;
  - `flexure/drainage_reorganisation.ipynb` follows where the next summer's water goes after a lake drains.
- **Variable thickness.** In 1-D this already works: pass `H` as an array. Both $D$ and $\Lambda$ scale as $H^3$, so $\tau$ is unchanged. The 2-D plates need a uniform thickness, because their moment potentials assume a uniform rigidity. Carrying the moment tensor instead, as MacAyeal et al. (2015) do, removes that restriction at about the same cost.
- **Lakes in map view.** This is implemented in `flexure/viscoelastic_plate.py`: the same physics with $\partial^4/\partial x^4 \to \nabla^4$ gives a viscoelastic plate. A round lake sinks less than a long trough at first, and the ring of tension survives. See Part 3 of `flexure/flexure_model_tutorial.ipynb`.
- **Nonlinear (Glen's law) creep and depth-varying temperature.** Here $\eta$ depends on stress and depth, so $\Lambda$ becomes a function of the state. The moment update becomes a local nonlinear update; the solve stays the same. MacAyeal et al. (2015, eqs 39–40) give a ready-made effective viscosity in terms of the bending moments. It is what keeps their lake basin from recovering after drainage (§5).
- **Membrane stresses and ice-shelf flow.** These are neglected, which is fine for bending over days to weeks. For coupled flow and flexure see MacAyeal et al. (2021).
- **Fracture.** The model reports stresses but does not break.

## 14 Parameters and references

| symbol | meaning | default |
|---|---|---|
| $H$ | ice thickness | 250 m |
| $E$ | effective Young's modulus (tidal-flexure studies infer about 1 GPa, well below the laboratory 9 GPa; e.g. Vaughan, 1995) | 1 GPa |
| $\nu$ | elastic Poisson's ratio | 0.3 |
| $\nu_v$ | Poisson's ratio of viscous creep (incompressible) | 0.5 |
| $\eta$ | ice viscosity | $10^{14}$ Pa s |
| $\rho_i$, $\rho_{sw}$, $\rho_w$ | ice, sea water, meltwater densities | 917, 1028, 1000 kg m$^{-3}$ |
| $D = EH^3/12(1-\nu^2)$ | flexural rigidity | $1.4 \times 10^{15}$ N m |
| $k = \rho_{sw} g$ | buoyancy stiffness | $1.0 \times 10^4$ Pa m$^{-1}$ |
| $\ell = (D/k)^{1/4}$ | flexural length | 614 m |
| $\tau = 4\eta(1-\nu^2)/E$ | mean Maxwell time | 4.2 days |
| $\tau_1 = 4\eta(1+\nu_v)(1-\nu)/E$ | relaxation time, isotropic part | 4.9 days |
| $\tau_2 = 4\eta(1-\nu_v)(1+\nu)/E$ | relaxation time, deviatoric part | 3.0 days |

The values of $E$ and especially $\eta$ are uncertain by an order of magnitude or more. Treat the numbers above as illustrative. Because $\ell \propto E^{1/4} H^{3/4}$ and $\tau \propto \eta / E$, it is easy to rescale the results.


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